

# Development of a Defect Estimation Method for CFRP Interstage Structures with Holes using Finite Element Analysis and Graph Neural Networks

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1. Motivation & measurement — defects in CFRP, infrared stress (DSPSS)
2. Method — difference-normalized input & a GAT on the FEM mesh
3. Results — 19-class 3-D localization (macro-F1 61%)
4. New: robustness to measurement noise (defect-free negatives)
5. Outlook

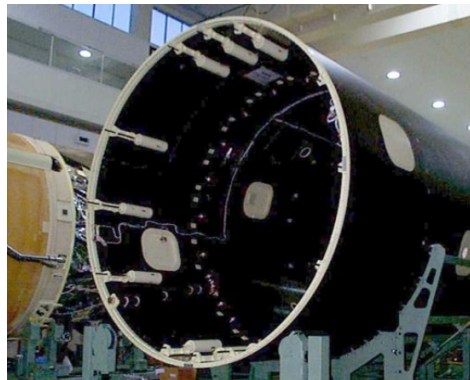
## Introduction: CFRP in space transportation

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CFRP carries primary load in reusable launch vehicles (high specific strength) [1, 2].

**Interlaminar delamination** around bolt holes critically degrades integrity, especially under repeated use. Reliable *defect localization* is essential.

Conventional NDT — ultrasonic [3], X-ray CT [4], tap testing [5] — is costly and operator-dependent [6].



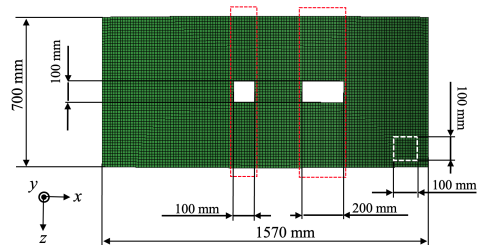
CFRP interstage structure of a launch vehicle.

## Infrared stress measurement (DSPSS)

Thermoelastic effect relates a small temperature change to the change in the sum of principal stresses:

$$\Delta T = -k T \Delta \sigma_{\text{sum}}, \quad \sigma_{\text{sum}} = \sigma_1 + \sigma_2 + \sigma_3 = \text{tr}(\sigma).$$

Infrared stress measurement [7–9] thus yields a **full-field surface map** of  $\sigma_{\text{sum}}$  (DSPSS), from which sub-surface defects are inferred.



DSPSS shows a stress valley over the defect.

## Prior work & problem formulation

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**Prior work.** Kojima *et al.* [10, 11] — a CNN *proof-of-concept* on *non-perforated* CFRP from FEM DSPSS.

**This work — harder setting:** *perforated* interstage  $\times$  3-D location (region  $\times$  layer)  $\times$  extreme imbalance ( $< 3\%$  defect nodes).

**Problem** — node classification on the mesh graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ , node features  $\mathbf{x}_i = (x, y, z, \hat{\sigma}_{\text{sum}})_i$ ; learn  $f_\theta : \mathcal{G} \rightarrow \mathbb{R}^{|\mathcal{V}| \times 19}$ ,

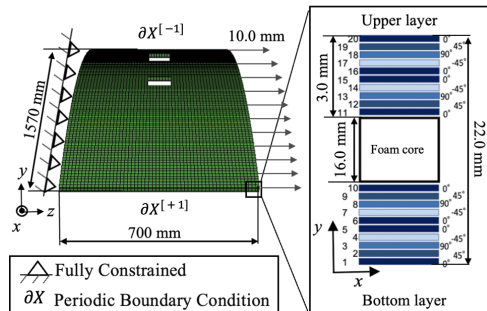
$$\hat{y}_i = \arg \max_{c \in \{0, \dots, 18\}} [\text{softmax } f_\theta(\mathcal{G})]_{i,c}, \quad \theta^\star = \arg \min_{\theta} \sum_{i \in \mathcal{V}} \text{FL}(p_{i,y_i}),$$

class 0 = defect-free; classes 1–18 = (in-plane region  $\times$  insertion layer).

## FEM analysis conditions

- 1/80-scale curved panel ( $R \approx 2$  m,  $L \approx 1.57$  m,  $t = 22$  mm).
- Holes of  $100 \times 100$  and  $200 \times 100$  mm.
- Sandwich laminate; tensile loading.
- Interlaminar delamination modeled explicitly by **Teflon inserts** between selected plies.

**Dataset:** 1 reference + **1386** defect specimens;  
5-fold CV ( $\approx 1110$  train / 278 test per fold).



FEM boundary conditions & hole geometry.

## Input data: normalized / difference DSPSS

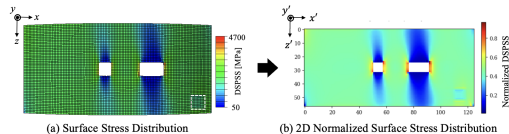
Raw DSPSS is dominated by **stress concentration around the hole**, masking the defect signal.

Remedy (paper):

- **Difference** field: damaged – defect-free reference;
- **Z-score normalization** (per node), improving depth discrimination.

$$d_i = \sigma_i^{\text{dmg}} - \sigma_i^{\text{ref}}, \quad \hat{\sigma}_i = \frac{d_i - \mu_d}{s_d}.$$

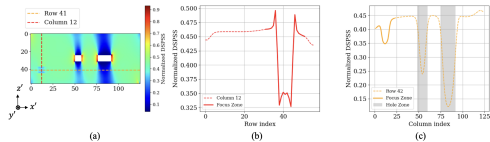
⇒ robust localization even near the opening.



Normalized DSPSS isolates the defect.

## Stress distribution around the defect

- DSPSS exhibits a clear **stress valley** at the defect centre.
- Depth and width of the valley depend on **ply angle**:  $90^\circ$  plies give sharp narrow valleys;  $0^\circ/45^\circ$  broaden them.
- This signature tells the network *which layer* the delamination sits in.



Normalized DSPSS profile across the defect.

## Implication

Layer discrimination is a *fine spatial-frequency* problem — motivating local attention.



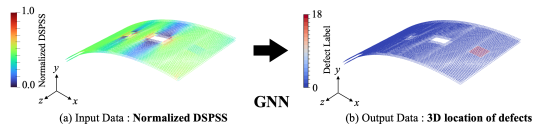
# Graph representation & GAT model

Nodes = elements ( $\mathbf{x}_i$ ), edges = fixed mesh connectivity. GAT message passing [12–14]:

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]),$$

$$\alpha_{ij} = \frac{\exp e_{ij}}{\sum_{k \in \mathcal{N}(i)} \exp e_{ik}},$$
$$\mathbf{h}'_i = \left\|_{m=1}^M \sigma \left( \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(m)} \mathbf{W}^{(m)} \mathbf{h}_j \right) \right\|.$$

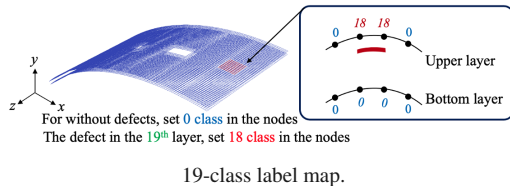
3 layers  $64 \rightarrow 256 \rightarrow 256$ ,  $M$  heads, BN + dropout + residual.



Graph model on the perforated mesh.

## Output: 3-D defect location as 19-class node labels

- Node classification into **19 classes**: class 0 = defect-free; remaining classes encode *in-plane region*  $\times$  *insertion layer*.
- One-side labelling so prediction needs only single-surface DSPSS.



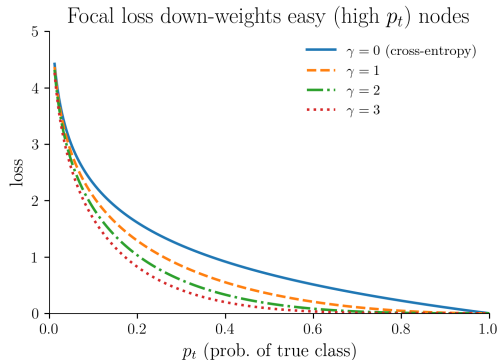
### Severe imbalance

Defect nodes are < 3% of all nodes — handled by the loss (next slide).

## Loss function: Focal loss

$$\text{FL}(p_t) = -\alpha_t (1 - p_t)^\gamma \log p_t.$$

- Down-weights easy, abundant defect-free nodes; emphasises rare defect-class nodes [15].
- Crucial because intact regions dominate (> 97%).



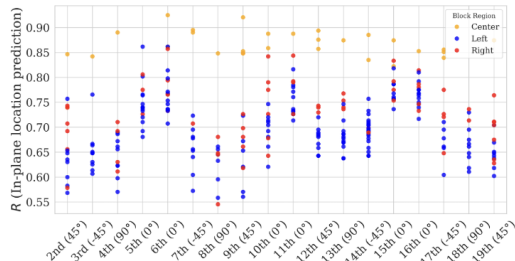
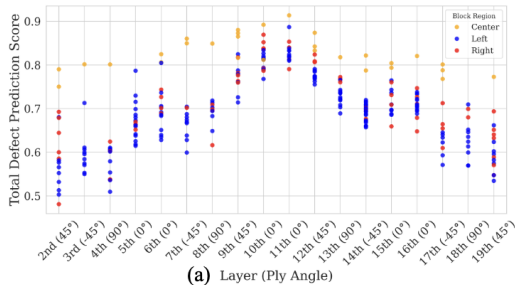
## Results: prediction accuracy

Test-set metrics:

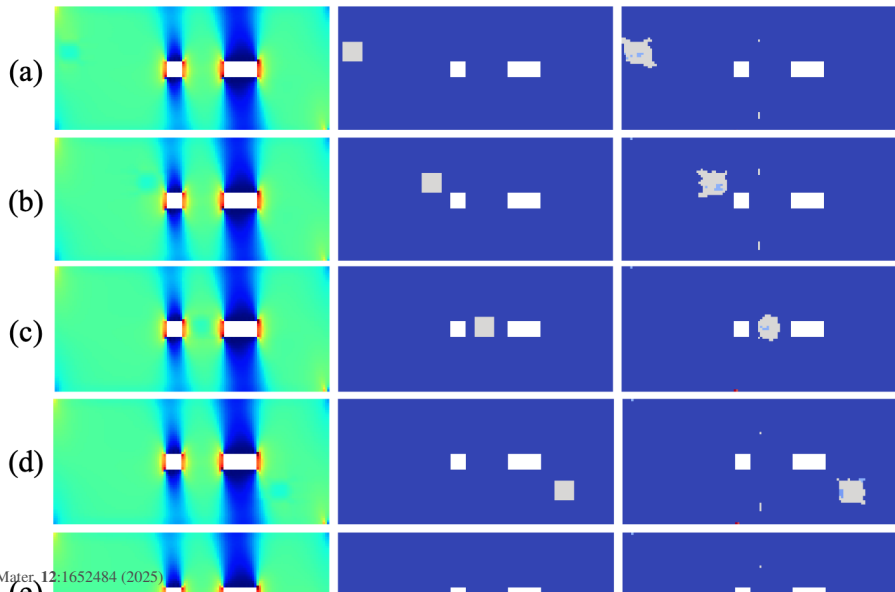
- planar  $R = 0.72$  (mean), depth  $P_{\text{gauss}} = 0.69$ , combined TDPS = 0.70 (best 0.92).
- Best layers: 9–12 (mid-thickness); Layer-10 samples all TDPS  $\geq 0.82$ .
- False-negative rate for defect nodes  $< 3\%$ .

$R$ : planar F1.  $P_{\text{gauss}}$ : Gaussian depth score ( $\sigma=2$ ).

$$\text{TDPS} = \frac{1}{2}(R + P_{\text{gauss}}).$$



## Qualitative predictions

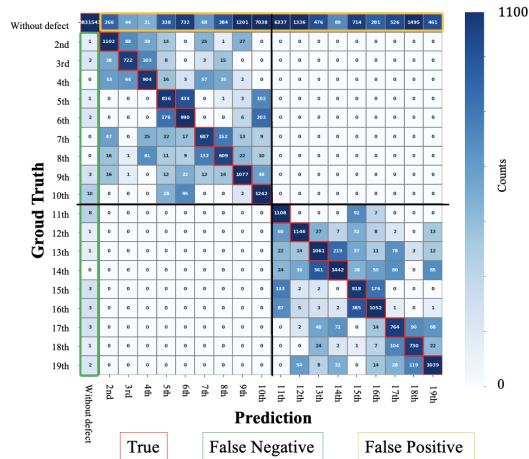


# Overall performance & confusion

## Headline

**Macro-F1 = 61%** across 19 classes, using *surface DSPSS only*.

- Dominant diagonal; residual *adjacent-layer* confusion.
- False positives cluster near the hole edge.
- Robust across defect insertion regions.



Confusion matrix (macro-F1 61%).

## Conclusion

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- Introduced an **FEM + GAT** framework for 3-D delamination localization in perforated CFRP interstage panels.
- **Difference-normalized DSPSS** suppresses hole stress concentration and enables full-region estimation.
- Achieved **macro-F1 61%** on 19 classes from surface data only, with a very low defect false-negative rate and no false detections in defect-free regions.

### A scalable foundation

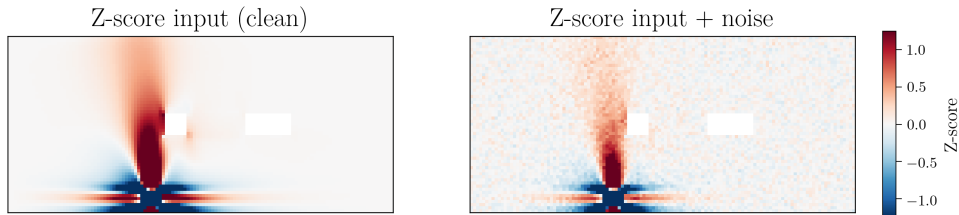
toward experimental infrared-stress (TSA) data and practical SHM of aerospace composite structures.

## New focus: robustness for real operation

The published model assumes clean FEM fields.  
In practice infrared measurement carries **acquisition noise**, and almost the whole structure is **defect-free** — false alarms are costly.

### WCCM contribution

Robustness under realistic noise *and* explicit suppression of false detection on healthy regions.



Clean vs. noise-injected Z-score input — the defect signature survives.



## Noise-augmented dataset & defect-free negatives (NDF)

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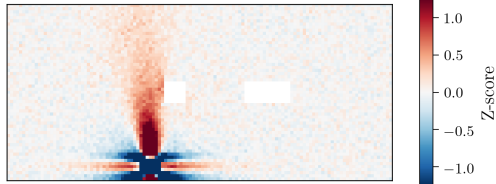
**Defect-free negatives (NDF):** zero field + Gaussian noise (10% of data std), 5,000 samples added to training.

Defect-free + 10% noise



**Structured noise on defect data:** line noise along the fibre ( $z$ ), attenuated noise in  $x$ , and global white noise.

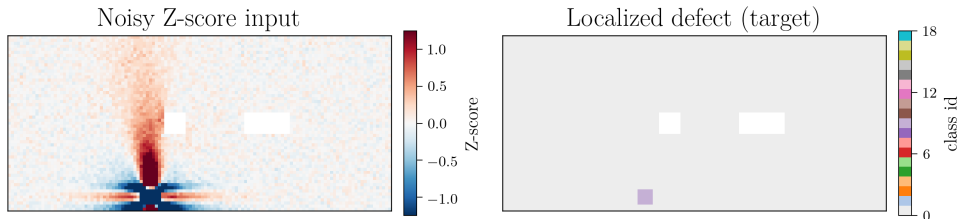
Defect + structured noise



Left: defect-free + noise (teaches “no defect”). Right: defect + structured noise.

## Results under noise

- Difference-DSPSS input enables **full-region** defect estimation; prediction stays accurate on **noise-injected** data.
- **No false detection** on defect-free inputs — the NDF negatives remove spurious activations.  $\Rightarrow$  robust enough to move toward *experimental* infrared stress data.



Noisy input (left)  $\rightarrow$  correct defect localization (right).

- **Sim-to-real:** apply the noise-robust model to *experimental* infrared-stress (TSA) measurements.
- **Architecture study (preliminary):** extending from attention GNNs to *mesh-physics* GNNs (MeshGraphNet) and layered variants; early results are encouraging and under evaluation with co-authors.
- **Application:** transfer to aerospace payload-fairing SHM.

*Preliminary; not part of the published results.*

## Acknowledgements

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- Conducted in collaboration with **JAXA** (Research & Development Directorate, Research Unit IV).
- Co-authors at **Keio University**, **Nagoya University**, and **JAXA**.
- Built on the FEA-generated CFRP interstage dataset of the prior study [10].

Thank you for your attention — questions welcome.

# References I

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- [1] Y. Hou et al. "On the damage mechanism of high-speed ballast impact and compression after impact for CFRP laminates". In: *Composite Structures* 229 (2019), p. 111435. doi: 10.1016/j.compstruct.2019.111435.
- [2] D. Kiefel et al. "Quantitative impact characterization of aeronautical CFRP materials with non-destructive testing methods". In: *AIP Conference Proceedings* 1650 (2015), pp. 591–598. doi: 10.1063/1.4914658.
- [3] S. Joas et al. "CFRP pipe inspection by means of air-coupled ultrasound". In: *AIP Conference Proceedings* 2055 (2019), p. 120003. doi: 10.1063/1.5084893.
- [4] M. Sultan et al. "On impact damage detection and quantification for CFRP laminates using structural response data only". In: *Mechanical Systems and Signal Processing* 25.8 (2011), pp. 3135–3152. doi: 10.1016/j.ymssp.2011.05.028.
- [5] P. Cawley and R. Adams. "An Automated Coin-Tap Technique for the Non-Destructive Testing of Composite Structures". In: *Non-Destructive Testing of Fibre Reinforced Plastics Composites*. Elsevier, 1987, pp. 11–15. doi: 10.1007/BF00614806.
- [6] A. Ngo et al. "Nondestructive Evaluation of Defects in Carbon Fiber Reinforced Polymer (CFRP) Composites". In: *Proceedings of SPIE – Smart Structures and Nondestructive Evaluation for Health Monitoring and Diagnostics*. Vol. 10169. 6 pages. SPIE, Apr. 2017, 101691P. doi: 10.1117/12.2260331.
- [7] T. Sakagami and S. Kubo. "Applications of pulse heating thermography and lock-in thermography to quantitative nondestructive evaluations". In: *Infrared Physics & Technology* 43.3–5 (2002), pp. 211–218. doi: 10.1016/S1350-4495(02)00141-X.
- [8] S. Keo et al. "Defect detection in CFRP by infrared thermography with CO2 Laser excitation compared to conventional lock-in infrared thermography". In: *Composites Part B: Engineering* 69 (2015), pp. 1–5. doi: 10.1016/j.compositesb.2014.09.018.
- [9] T. Sakagami et al. "Application of infrared stress analysis for damage detection in bridge structures". In: *Procedia Structural Integrity* 2 (2016), pp. 2132–2139. doi: 10.1016/j.prostr.2016.06.265.
- [10] Y. Kojima et al. "Inverse estimation method for internal defects based on surface stress of carbon-fiber-reinforced plastics using machine learning". In: *Advanced Composite Materials* 31.6 (2022). Published online: 30 Mar 2022, pp. 617–629. doi: 10.1080/09243046.2022.2052786.
- [11] Y. Kojima et al. "Transfer-learning-aided Defect Prediction in Simply Shaped CFRP Specimens Based on Stress Distribution Obtained from Finite Element Analysis and Infrared Stress Measurement". In: *Composites Part B: Engineering* 291 (2024), p. 111958. doi: 10.1016/j.compositesb.2023.111958.

## References II

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- [12] P. Veličković et al. “Graph Attention Networks”. In: *International Conference on Learning Representations (ICLR) 2018*. 2018. doi: 10.48550/ARXIV.1710.10903. URL: <https://arxiv.org/abs/1710.10903>.
- [13] F. Scarselli et al. “The Graph Neural Network Model”. In: *IEEE Transactions on Neural Networks* 20.1 (2009), pp. 61–80. doi: 10.1109/TNN.2008.2005605.
- [14] J. Gilmer et al. “Neural Message Passing for Quantum Chemistry”. In: *Proceedings of the 34th International Conference on Machine Learning (ICML)*. Vol. 70. Proceedings of Machine Learning Research. PMLR, 2017, pp. 1263–1272. doi: 10.48550/arXiv:1704.01212.
- [15] T. Lin et al. “Focal Loss for Dense Object Detection”. In: *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. IEEE, 2017, pp. 2980–2988. doi: 10.1109/ICCV.2017.324.

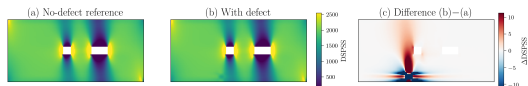
# Appendix

## Appendix: difference-data construction

Raw DSPSS near a hole is dominated by geometric stress concentration. We remove it by a per-specimen **difference**:

$$\text{Difference} = \underbrace{\text{Original (defect)}}_{\text{damaged}} - \underbrace{\text{Reference (no defect)}}_{\text{pristine}}.$$

The residual isolates the defect-induced perturbation, independent of the hole's baseline field.



With-defect – no-defect reference = difference field (defect localized).

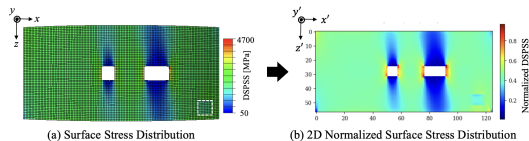


## Appendix: Z-score normalization

The difference field is standardized per node (Z-score):

$$\hat{\sigma} = \frac{\sigma - \mu}{s}.$$

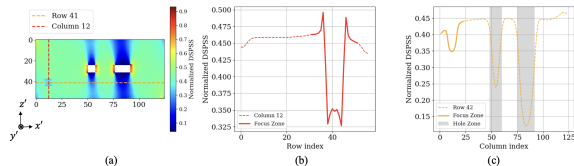
- equalizes magnitude across regions/specimens;
- improves **depth-direction** (layer) discrimination;
- keeps the network NDT-admissible (coordinate/stress only).



Normalized DSPSS input (paper figure).

## Appendix: DSPSS profile around the defect

The difference-DSPSS shows a pronounced **stress valley** at the defect centre; its depth and width encode the insertion layer (ply-angle dependence).



Profile of the normalized DSPSS across the defect (paper figure).

## Appendix: evaluation metrics

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- $R$  — in-plane (planar) F1 score for defect vs. non-defect at the correct location.
- $P_{\text{gauss}}$  — depth-direction score: Gaussian-weighted ( $\sigma = 2$  layers) credit for predicting the correct or a near-correct insertion layer.
- **TDPS** (Total Defect Prediction Score) =  $\frac{1}{2} (R + P_{\text{gauss}})$ .
- **macro-F1** — unweighted mean F1 over all 19 classes (the headline metric).

These quantify in-plane localization ( $R$ ), depth/layer accuracy ( $P_{\text{gauss}}$ ), and their combination (TDPS), complementing the class-level macro-F1.